Haniel Reena D R

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EDUCATION

Institute of Science & Technology, NMV University — Chennai, India
Bachelor of Technology in Computer Science Engineering *Expected May - 2027*
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Computer Networks, Operating Systems, Machine Learning, Artificial Intelligence, Computer Vision, Natural Language Processing.
CGPA: 9.1/10

TECHNICAL SKILLS

Deep Learning & AI: Deep Learning, ResNet-18, Multimodal AI, Anomaly Detection, 3-NN, Memory-Bank Methods
NLP & Language Models: BART, T5, Tesseract OCR, Abstractive Summarization, Clinical Text Processing, ROUGE Evaluation
Computer Vision & Medical Imaging: OpenCV, Image Processing, Face Detection, Medical Imaging, 3D Volumetric Segmentation, Morphological Processing, Otsu Thresholding, Gaussian Filtering
Data Science & Machine Learning: Python, NumPy, scikit-learn, Random Forest, Logistic Regression, Data Preprocessing, Model Training, Multi-Class Classification
Programming & Development: HTML, SQL, API Integration, Database Connectivity, Git, GitHub, Visual Studio Code, Jupyter Notebook

PROJECT EXPERIENCE

Adaptive Multimodal Open-World AI / Anomaly Detection | *Python, Deep Learning, Multimodal AI*
- Designed an RGB–XYZ multimodal anomaly-detection framework using multiscale ResNet-18 features, normal memory banks, and 3-NN patch-level anomaly scoring.
- Implemented pseudo-anomaly validation and modality-aware fusion, achieving 85.77% AUROC, 75.86% F1-score, 94.83% precision, and 63.22% recall on MVTec 3D-AD.

3D Kidney Medical Imaging & Visualization | *Python, NumPy, OrganMNIST3D, Plotly*
- Developed a 3D kidney imaging pipeline for volumetric preprocessing, segmentation, localization, and interactive visualization.
- Applied Gaussian filtering, Otsu thresholding, morphological processing, and volumetric segmentation, achieving 85.7% Dice score, 78.4% IoU, and 90.6% sensitivity.

Intelligent Clinical Companion / Medical Report Summarizer | *Python, OCR, NLP, BART, T5*
- Developed an AI medical-report system for OCR-based extraction, summarization, and patient-friendly text generation.
- Integrated Tesseract OCR, BART, and T5, achieving 46.2% ROUGE-1, 28.5% ROUGE-2, and 41.3% ROUGE-L.

Network Intrusion Detection & Explainable AI | *Python, Scikit-learn, Random Forest, UNSW-NB15*
- Developed a multi-class intrusion detection system using the UNSW-NB15 dataset across 10 normal/attack traffic categories.
- Compared Logistic Regression and Random Forest with preprocessing and multi-class evaluation; Random Forest achieved 86.23% accuracy and 57.52% Macro-F1 on the official 82,332-record test set.

Real-Time Age & Gender Detection System | *Python, OpenCV, Deep Learning*
- Developed a real-time computer-vision pipeline for face detection, age estimation, and gender classification from webcam streams.
- Implemented image preprocessing, face-region extraction, deep-learning inference, and bounding-box visualization, achieving 92.3% Gender-Classification Accuracy, 5.1-Year Age MAE, and 24 FPS during real-time inference.

PUBLICATIONS

High-Volume Electronics Manufacturing with Automated X-Ray Inspections *Accepted for Publication, 2026*
- Developed an imbalance-aware deep learning pipeline for automated X-ray inspection using ResNet-18, transfer learning, decision-threshold optimization, and false-call reduction analysis.
- Addressed a 91:1 class imbalance using Youden Index-based threshold optimization to preserve defect sensitivity while reducing false positives.
- Achieved 89.1% Weighted F1-Score, 86.7% Balanced Accuracy, 91.6% Sensitivity, 0.9221 AUROC, and 81.8% False-Call Reduction in the reported evaluation.

PROFESSIONAL EXPERIENCE

Inertz Technologies | Full Stack Intern | Chennai, India *January 2025 — June 2025*
- Developed frontend and backend modules using HTML, Node.js, and SQL, integrating web application components with database connectivity and API functionality.
- Implemented database connectivity and API integration to support application features, while collaborating with developers to test, debug, and optimize application modules.
- Assisted in end-to-end web application development, performing functional testing, bug fixing, and backend integration to improve application reliability and performance.